Daffodil International University

Faculty of Science & Information Technology

Department of Computer Science & Engineering

Retake Examination, Spring 2026

Course Code: CSE321, Course Title: System Analysis and Design

Section: RE_A(3C)

Time: 01:30 Hrs	Marks: 25

Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

Greenfield City Hospital currently handles its patient appointment and record management process manually. Patients visit the hospital in person to collect and fill paper-based appointment forms, pay consultation fees at separate billing counters, and wait in long queues to check appointment status. Hospital staff manually verify patient eligibility for specialist referrals, record medical data in spreadsheets, and post daily appointment schedules on notice boards. This results in long waiting times, frequent data entry errors, misplaced medical records, and delayed treatment scheduling. The hospital has decided to develop a Smart Online Hospital Management System (SOHMS). The system will allow patients to register and book appointments online, upload required medical documents, pay consultation fees electronically, and track their appointment and treatment status in real time. The system will automatically verify patient eligibility for specialist consultations based on referral and health condition rules, generate daily and weekly appointment schedules, assign patients to appropriate departments and doctors, and produce monthly health service reports for hospital administration.

Now answer the following questions

Q. No.	Question	Marks	CO
1.	Answer the following based on Data, Information and Management Structure: **a)** Identify and explain the Operational, Tactical, and Strategic information required for the SOHMS. Clearly describe how each type of information relates to different levels of the hospital management structure. **b)** Explain how good interpersonal relations, effective communication skills, and analytical thinking ability of a System Analyst contribute to the successful development of the SOHMS. Provide brief examples from the hospital management context.	**[4]**	**CO1**
2.	List and explain all nine steps of the System Analysis and Design Life Cycle. For each step, describe ONE specific task that would be performed during the development of the SOHMS.	**[5]**	**CO1**
3.	A system analyst visited the Greenfield City Hospital to understand the current appointment process. The analyst conducted the following session with the Head of Patient Services: **Analyst:** *"Can you describe how a patient's appointment is processed from booking to final scheduling?"* **Head:** *"The patient fills in a paper form, we manually check the referral letter and medical documents, enter patient details and doctor availability into a spreadsheet, then assign a slot. During peak seasons, it often takes two to three days."* **Analyst:** *"What are the major problems you face with this current process?"* **Head:** *"Medical records frequently go missing, data entry mistakes are common, and patients constantly call to check if their appointment has been confirmed."* **Analyst:** *"What would you want the new system to do that the current one cannot?"* **Head:** *"Automate eligibility and referral checking, show real-time appointment status for patients, and generate the daily schedule automatically."* i. Identify the information gathering technique used above and justify your answer. ii. Identify different sources of information for gathering the system requirements of SOHMS. Specify an appropriate information-gathering technique for each source and justify your answer. iii. If a survey form is required to collect information from patients, design a suitable sample form.	**[5]**	**CO2**

4.	Draw the Level 0 (Context Diagram) and Level 1 Data Flow Diagram (DFD) for the SOHMS. Clearly show the External Entities, Major Processes, Data Stores and Data Flows in your diagrams. Ensure that all elements are properly labelled and logically connected.	**[2+4]**	**CO2**
5.	In SOHMS, when a patient submits an appointment request for a specialist department, the system follows these rules: The system first checks whether a valid referral letter has been uploaded. If the referral letter is missing, the appointment request is rejected. If the referral letter is present, the system checks the patient's National Health Insurance (NHI) status. If the patient is not NHI registered, the appointment is marked as SELF-FUNDED and moved to a payment queue. If the patient is NHI registered, the system checks whether the requested specialist falls within the approved NHI coverage. If the specialist is not covered, the appointment is marked as SELF-FUNDED. If the specialist is covered by NHI, the appointment is marked APPROVED and added to the scheduling queue. However, if the patient has been flagged as an EMERGENCY CASE by a physician, the system bypasses all referral and NHI checks and immediately marks the appointment as EMERGENCY APPROVED. **a)** Draw a Decision Tree representing the appointment eligibility verification logic of the system. **b)** Construct a Decision Table based on the decision tree developed.	**[2+3]**	**CO2**
6.	Based on the SOHMS scenario above, identify the deficiencies of the current manual hospital management system and set goals to remove those deficiencies. List any FOUR deficiencies found in the current system and classify each as: Missing Function, Unsatisfactory Performance, or Excessive Cost of Operations. Then formulate TWO Main Goals (M1, M2) and TWO Sub-Goals under each Main Goal (S1.1, S1.2 under M1 and S2.1, S2.2 under M2). Each goal must be quantified and specific. Finally, propose THREE alternative solutions to meet these goals and recommend the best one with justification.	**[6]**	**CO2**

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